CSS4005 Final Project — Grading Rubric (40 points)

Part	Points
Report (graded with the table below)	**20**
Presentation & demo	**8**
Oral defense	**12**
Total	**40**

Each report area is scored at three levels:

- **Full**: everything in the README checklist is done **and** shown with evidence, and your variant values are applied.
- **Partial** (about half): built but evidence is missing or weak, some checklist items are missing, or variant values are not applied.
- **None** (0): missing, not working, or only described with no evidence.

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1. Report — 20 points

#	Area	README §	Points	"Full" means
R1	Variant applied + architecture (diagram, shared responsibility, ≥ 4 ADRs)	4, 5.8	2	Variant table complete with evidence; diagram matches what was built; ADRs give real trade-offs
R2	Networking	5.1	2	3 tiers × 2 zones from your CIDR, SG/firewall chaining by reference, second layer, DB unreachable test
R3	Compute, Linux hardening	5.2	1	Templates + startup scripts for nodes and DB, non-root, no public SSH, no keys in user data
R4	**Kubernetes cluster**	5.4	3	Cluster per your `cluster` variant over 2 zones; Deployment with requests/limits + both probes; zone spreading; Service + Ingress behind the cloud LB; HPA at `hpa_cpu_target`; rolling update + rollback; node/zone drill with pods rescheduled
R5	Load balancing & auto scaling	5.5	2	LB in 2 zones with `/readyz` checks; pod + node scaling fit to `traffic_pattern`; load-test graphs; self-healing; HTTPS
R6	Storage	5.3	1	Versioning, lifecycle, BPA, encryption, signed URLs, separate DB disk, scheduled snapshots with variant retention
R7	**Self-hosted database**	5.6	3	Primary + replica in 2 zones (+ ClickHouse if required); 3 roles with denied actions; password policy rejecting weak passwords; backups meeting RPO; restore test and failover drill timed vs RTO; managed-vs-self-hosted comparison
R8	IAM & secrets	5.7	2	Roles/service accounts least privilege (or documented ideal policy on Learner Lab); all secrets in the cloud secret manager, fetched at boot/deploy; nothing secret in git/images/state outputs; rotation plan
R9	Containers & CI/CD	5.9	1	Lint/test/build/scan/push with immutable tags; deploy to the cluster gated by readiness; rollback demonstrated

#	Area	README §	Points	"Full" means
R10	Security checks	5.10	1	gitleaks, image scan, IaC scan, password tests run in CI; findings table complete with status
R11	Infrastructure as Code	5.11	1	Everything in Terraform modules (+ manifests/Helm); protected state; fmt/validate in CI; timed destroy → apply rebuild
R12	Monitoring, logging, HA drills & cost	5.12, 5.13	1	Dashboards incl. pods/DB; ≥ 4 alerts with variant thresholds, one firing → resolving; `request_id` log query; zone drill; cost estimate vs actual, budget alert at cap, ≥ 3 optimizations; teardown proof
	Total		**20**	

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2. Presentation & demo — 8 points

Criterion	Points
Clear architecture walk-through tied to **your** variant	2
Live demo (or demo video) of the working platform through the LB, incl. the cluster (`kubectl get pods -o wide` , HPA)	3
One drill shown live or on video (node/zone failure, DB failover or rollback)	2
Time kept (~7 min), readable slides/diagram	1

3. Oral defense — 12 points

Four questions, 3 points each (3 = correct and explained in your own words with reference to your build, 2 = mostly correct, 1 = vague, 0 = can't answer). At least one question is about a file in **your** repository you must explain line by line, and at least one is about the Kubernetes cluster or the self-hosted database.

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4. Penalties and caps

Situation	Effect
A secret (password, key, token) committed to the repo or baked into an image	–4 points and must be rotated before grading
Database reachable from the internet	Report capped at **10 / 20**
Managed DB (RDS / Cloud SQL) used instead of a self-hosted engine	R7 = 0 and report capped at **12 / 20**
Database placed inside the Kubernetes cluster	R7 capped at partial (1.5 / 3)
Variant ignored (wrong CIDR, prefix, engine, cluster type, thresholds)	Report capped at **12 / 20**
Cannot explain own code at the defense	Defense capped at **4 / 12**
Plagiarism (shared code/report, another student's platform)	0 for the whole project, reported per university policy
Resources left running after grading with no teardown proof	–2 points
Kubernetes cluster shown **only on a local kind / k3d cluster** because cloud credits ran out (see FREE_TIER_GUIDE.md §7)	R4 capped at **half (1.5 / 3)**; the cloud LB → cluster path in R5 counts as partial. Every other area still needs **cloud** evidence (screenshots, outputs) or the demo video recorded while it ran in the cloud

Documented **Learner Lab limitations** (no IAM role creation, blocked services, session expiry) are **not** penalised when the report explains the limitation and the real-account design.